I) General Information on the Dataset

Our project utilizes a hybrid dataset approach. The target variables (product reorder ratios and purchase frequencies) are derived from the tabular Kaggle "Instacart Market Basket Analysis" dataset. Because this tabular data lacks visual features, we are scraping the corresponding product packaging images directly from Instacart to serve as our input data.

Vision Tasks:

Our primary vision task is end-to-end image regression/classification. We will input the raw product packaging images into a Convolutional Neural Network (CNN) to directly predict the tabular reorder probability of that product.

II) Ethical Considerations

- **Scraping Ethics (Prompt II.a):** We are scraping data from Instacart. To mitigate ethical concerns regarding server disruption and terms of service, we are utilizing the Instacart API with local throttling and routing through a VPN/TOR network to minimize our footprint and avoid negatively impacting their infrastructure.
- **Dataset Bias:** There is a structural variance in our scraped images—most products are featured on clean, transparent backgrounds, while others are photographed in-store. This discrepancy could introduce bias, where the model learns to associate image formatting (e.g., lighting or background noise) with purchase behavior rather than the actual branding features.

III) Initial Exploratory Analysis of the Data

Exploration of the visual data reveals two primary image compositions: products isolated on transparent backgrounds and products photographed in a physical retail environment. The tabular Kaggle dataset will be joined to these images on a 1:1 basis using product names/IDs to establish the ground-truth reorder metrics.

IV) Thoughts and Initial Experimentation (Proving DL is Required)

Deep learning is strictly required for this project because traditional machine learning algorithms cannot natively process raw pixel data to find complex, non-linear correlations with consumer behavior. While standard ML works for the tabular data, extracting abstract branding features—such as typography, logo prominence, and color psychology—from images requires the hierarchical feature extraction capabilities of a CNN. A deep learning architecture will automatically learn these spatial hierarchies without the need for manual feature engineering, allowing us to map visual packaging cues directly to brand loyalty metrics.